AN ORDER ON THE LEECH LATTICE FROM TRANSPOSITION-TWISTED OCTONIONS

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Abstract. We show that the Leech lattice Λ admits the structure of an order in a 24-dimensional real algebra. The product is built from three identical copies of a *transposition-twisted* octonion algebra — the algebra obtained by applying a single transposition to the imaginary basis elements of the standard Fano-plane multiplication — with $\mathbb{Z}_3$-symmetric cross-block routing. Wilson's octonionic characterisation of Λ imposes three sublattice conditions. The standard (untwisted) triple-octonion product satisfies conditions 1 and 2 but fails condition 3; we prove symbolically that the twist corrects this failure by mapping the condition-3 sublattice to one that is closed under the standard octonion product. The construction is essentially unique: all twenty-one transpositions yield the same algebra up to Fano-plane automorphisms.

This result was developed through a systematic human–AI collaboration; the complete research record is publicly available [10].

1. Introduction

The Leech lattice Λ is the unique even unimodular lattice of rank 24 with no vectors of squared norm 2 [3]. Its minimal shell $\mathrm{Min}(\Lambda)$ consists of 196,560 vectors of squared norm 8, and its automorphism group is the Conway group Co_0, a double cover of the sporadic simple group Co_1.

In rank 8, the E_8 lattice L carries an octonionic multiplication that makes it a maximal order in the real octonion division algebra. There are seven such orders, one for each of the seven maximal orders of the integral octonions [5, 4]. Wilson [9] showed that Λ embeds naturally in L^3 via three sublattice conditions, raising the question: does Λ inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Dixon [6] constructed Λ using an octonionic XY-product but did not test multiplicative closure. Baez and Egan [2] explored Jordan-algebraic structures on Λ within the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, finding closure under a doubled Jordan product but not under the standard one. The present author participated in the 2014 discussion, testing Jordan products without finding closure on the Leech inner shell.

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The standard triple-octonion product on L^3 — using the same octonion multiplication in all three blocks with $\mathbb{Z}_3$-symmetric cross-block routing — satisfies Wilson's conditions 1 and 2 automatically but fails condition 3. The failure is structural: condition 3 requires $x+y+z \in Ls$ for the block sum, and the standard product does not preserve the sublattice Ls under multiplication.

The main result of this paper is that a single transposition applied to the Fano-plane triples corrects this failure. The key insight is that the twist maps the condition-3 sublattice Ls to $\sigma(Ls)$, a different sublattice of L that *is* closed under the standard octonion product.

Theorem 1.1. *Let $\cdot_\sigma$ denote the transposition-twisted octonion multiplication on $\mathbb{R}^8$ (Definition 3.1), and let $\star$ denote the associated triple product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ with $\mathbb{Z}_3$ cross-block routing (Definition 3.5). Then for all $u, v \in \Lambda$, the product $u \star v$ lies in Λ.*

Corollary 1.2. $(\Lambda, +, \star)$ *is an order in the* $\mathbb{R}$*-algebra* $(\mathbb{R}^{24}, +, \star)$.

Proof. $\mathrm{Min}(\Lambda)$ generates Λ over $\mathbb{Z}$ [3]. The product $\star$ is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1, $\star$ maps $\Lambda \times \Lambda$ into Λ. □

2. Preliminaries

2.1. **The octonion algebra.** Let $\mathbb{O}$ denote the real octonion algebra with basis $\{e_0, e_1, \ldots, e_7\}$, where e_0 is the identity. The multiplication is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{1}$$

encoding the rule: for each triple (a, b, c), $e_a \cdot e_b = +e_c$, with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [1]. Every imaginary unit squares to $-e_0$. The algebra $\mathbb{O}$ is the unique real normed division algebra of dimension 8 [1].

The *norm form* is $N(x) = x\bar{x} = \sum_{i=0}^{7} x_i^2$ for $x = \sum_i x_i e_i$, where $\bar{x} = x_0 e_0 - \sum_{i=1}^{7} x_i e_i$ is the conjugate. The composition property $N(xy) = N(x)N(y)$ holds [1].

2.2. **The E_8 lattice.** Wilson's E_8 lattice is $L = D_8^+$, the set of vectors in $\mathbb{R}^8$ whose coordinates (in the basis $\{e_0, \ldots, e_7\}$) are either all integers or all half-integers, with even coordinate sum [9, 4]. It has 240 roots (vectors of squared norm 2). Under the standard octonion multiplication, L is a maximal order in $\mathbb{O}$: it is closed under multiplication [5, 4].

2.3. **Wilson's Leech lattice.** Define $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$ and $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$. Both lie in L with $N(s) = N(\bar{s}) = 2$. Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \qquad Ls = \{\ell \cdot s : \ell \in L\}.$$

These are sublattices of L with $[L : L\bar{s}] = [L : Ls] = 16$, $L\bar{s} + Ls = L$, and $L\bar{s} \cap Ls = 2L$ [9].

Definition 2.1 (Wilson [9, Section 3])**.** The *Leech lattice* is

$$\Lambda = \big\{(x, y, z) \in L^3 : (1)\ x, y, z \in L;\ (2)\ x{+}y,\ x{+}z,\ y{+}z \in L\bar{s};\ (3)\ x{+}y{+}z \in Ls\big\}.$$

The 196,560 minimal vectors of Λ decompose into three families [9]:

Family	Formula	Count
Type 1	$(2\lambda, 0, 0)$ and cyclic permutations	$3 \times 240 = 720$
Type 2	$(\lambda\bar{s},\ (\lambda\bar{s})j,\ 0)$ and cyclic perms	$3 \times 240 \times 16 = 11{,}520$
Type 3	$((\lambda s)j,\ \lambda k,\ (\lambda j)k)$ and cyclic perms	$3 \times 240 \times 16^2 = 184{,}320$

Here λ ranges over the 240 roots of L, and $j, k \in J = \{\pm e_t : 0 \le t \le 7\}$ (16 unit octonions).

3. The construction

Definition 3.1 (Transposition-twisted octonion algebra)**.** Let $\sigma = (s\ t)$ be a transposition on $\{1, \dots, 7\}$. Extend σ to a linear map $\sigma : \mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and permutes $e_s \leftrightarrow e_t$. Apply σ to each standard Fano triple: $(a, b, c) \mapsto (\sigma(a), \sigma(b), \sigma(c))$. The resulting triples define a new multiplication table on $\mathbb{R}^8$. Equivalently, the *σ-twisted multiplication* is

$$x \cdot_\sigma y \;=\; \sigma\big(\sigma(x) \cdot \sigma(y)\big), \tag{2}$$

where $\cdot$ denotes the standard octonion product.

Proposition 3.2. *$(\mathbb{R}^8, \cdot_\sigma)$ is an octonion algebra, isomorphic to $(\mathbb{O}, \cdot)$ via the map σ itself:*

$$\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y) \qquad \textit{for all } x, y \in \mathbb{R}^8. \tag{3}$$

In particular, $\cdot_\sigma$ satisfies the composition property $N(x \cdot_\sigma y) = N(x)N(y)$.

Proof. For any $x, y \in \mathbb{R}^8$:

$$\sigma(x) \cdot_\sigma \sigma(y) = \sigma\big(\sigma(\sigma(x)) \cdot \sigma(\sigma(y))\big) = \sigma(x \cdot y),$$

since $\sigma^2 = \mathrm{id}$. Thus $\sigma : (\mathbb{O}, \cdot) \to (\mathbb{R}^8, \cdot_\sigma)$ is a vector-space isomorphism that preserves multiplication. Composition follows. □

Remark 3.3. For the transposition $\sigma = (1\ 2)$, exactly 30 of the 64 multiplication table entries change relative to the standard table: 6 entries undergo a sign change only, 16 undergo a target-index change only, and 8 undergo both. Despite the large number of differences, the algebra is isomorphic to $\mathbb{O}$.

Remark 3.4. The 21 transpositions of $\{1, \dots, 7\}$ form a single orbit under the Fano-plane automorphism group $\mathrm{GL}(3, \mathbb{F}_2)$ (which has order 168 and acts on the 7 points of the Fano plane). Therefore all transposition twists produce the same multiplication table up to basis relabelling. The construction is essentially unique.

Definition 3.5 (Triple product with $\mathbb{Z}_3$ routing)**.** Decompose $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ into three copies of $\mathbb{R}^8$ (blocks indexed $\alpha = 1, 2, 3$). For $u = (x, y, z)$ and $v = (x', y', z')$ with $x, x' \in \mathbb{O}_1$, etc., define the *triple product*

$$u \star v = (P_1,\, P_2,\, P_3), \tag{4}$$

where each block P_α is the sum of three σ-twisted octonion products:

$$P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z', \tag{5}$$

$$P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x', \tag{6}$$

$$P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'. \tag{7}$$

The routing rule is: a product of blocks α and β lands in block γ, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ for $\alpha \neq \beta$, and in block α when $\alpha = \beta$. All nine block-pair products use the single multiplication $\cdot_\sigma$.

4. Proof of closure

We prove Theorem 1.1 by showing that the twisted product preserves each of Wilson's three conditions. The proof rests on five lemmas about the interaction between the transposition σ, the maximal order L, and the sublattices Ls and $L\bar{s}$.

4.1. Reduction to standard products.

Since each σ-twisted product has the form $a \cdot_\sigma b = \sigma(\sigma(a) \cdot \sigma(b))$, and σ is linear, each block P_α is σ applied to a sum of standard octonion products of σ-images of the inputs.

Write $X = \sigma(x)$, $Y = \sigma(y)$, $Z = \sigma(z)$, $X' = \sigma(x')$, $Y' = \sigma(y')$, $Z' = \sigma(z')$. Then:

$$P_1 = \sigma\big(X \cdot X' + Z \cdot Y' + Y \cdot Z'\big), \tag{8}$$

$$P_2 = \sigma\big(Y \cdot Y' + X \cdot Z' + Z \cdot X'\big), \tag{9}$$

$$P_3 = \sigma\big(Z \cdot Z' + Y \cdot X' + X \cdot Y'\big). \tag{10}$$

Taking sums:

- **Block sum.** $P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big)$.
- **Pairwise sums.** For example, $P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big)$, and similarly for the other two pairs.

4.2. The five lemmas.

Lemma 4.1 (Lemma A)**.** *$\sigma(L) = L$.*

Proof. The lattice $L = D_8^+$ consists of all vectors in $\mathbb{R}^8$ whose coordinates are either all integers or all half-integers, with even coordinate sum. The transposition σ permutes two coordinates, which preserves both the integer/half-integer parity and the coordinate sum. Hence $\sigma(L) \subseteq L$. Since σ is an involution, $\sigma(L) = L$. $\square$

Lemma 4.2 (Lemma B)**.** *$L \cdot L \subseteq L$.*

Proof. L is a maximal order in $\mathbb{O}$ [5, 4]. □

Lemma 4.3 (Lemma C)**.** *$L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$.*

Proof. Both L and $\sigma(L\bar{s})$ are lattices of rank 8. It suffices to check the inclusion on $\mathbb{Z}$-bases: if $\{b_1, \ldots, b_8\}$ is a $\mathbb{Z}$-basis of L and $\{c_1, \ldots, c_8\}$ is a $\mathbb{Z}$-basis of $\sigma(L\bar{s})$, then $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$ holds if and only if each of the 64 products $b_i \cdot c_j$ is an integer combination of $\{c_1, \ldots, c_8\}$. This is verified by exact arithmetic (using rational numbers, no floating point) on explicit bases. All 64 products lie in $\sigma(L\bar{s})$. □

Lemma 4.4 (Lemma D)**.** *$\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.*

Proof. As in Lemma C, verified on $\mathbb{Z}$-bases by exact arithmetic: all 64 products of basis vectors of $\sigma(Ls)$ lie in $\sigma(Ls)$. □

Lemma 4.5 (Lemma E)**.** *$\sigma(Ls) \neq Ls$.*

Proof. Let $\sigma = (1\ 2)$. Consider the element $v = (-1, 0, 1, 0, 0, 1, 0, 1)$, which lies in $\sigma(Ls)$. Direct computation shows that v is not an integer combination of any $\mathbb{Z}$-basis for Ls, so $v \notin Ls$. □

Remark 4.6. The sublattice Ls is *not* closed under the standard octonion product: there exist basis vectors $a, b \in Ls$ with $a \cdot b \notin Ls$. This is the reason the untwisted triple product fails condition 3. The twist maps the condition-3 sublattice from Ls (where closure fails) to $\sigma(Ls)$ (where closure holds, by Lemma D).

4.3. **Condition 1: each block lies in L.**

Proposition 4.7. *For all $u, v \in \Lambda$, each block P_α of $u \star v$ lies in L.*

Proof. Consider $P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z')$. Since $u \in \Lambda$, condition 1 gives $x \in L$, so $X = \sigma(x) \in L$ by Lemma A. Similarly $X', Y, Y', Z, Z' \in L$. By Lemma B, each product $X \cdot X'$, $Z \cdot Y'$, $Y \cdot Z'$ lies in L. Their sum lies in L (closed under addition). Applying σ and using Lemma A, $P_1 \in L$. The same argument applies to P_2 and P_3. □

4.4. **Condition 2: pairwise sums lie in $L\bar{s}$.**

Proposition 4.8. *For all $u, v \in \Lambda$, the pairwise sums $P_\alpha + P_\beta$ lie in $L\bar{s}$.*

Proof. We show $P_1 + P_2 \in L\bar{s}$; the other two cases follow by the $\mathbb{Z}_3$ symmetry of the routing.

From (**??**)–(**??**):

$$P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big).$$

Since $v \in \Lambda$, condition 2 gives $x' + z' \in L\bar{s}$, so $X' + Z' = \sigma(x' + z') \in \sigma(L\bar{s})$. Similarly $Y' + Z' \in \sigma(L\bar{s})$ and $X' + Y' \in \sigma(L\bar{s})$. Since $u \in \Lambda$, we have $X, Y, Z \in L$ (by Lemma A).

By Lemma C, each product $X \cdot (X' + Z')$, $Y \cdot (Y' + Z')$, $Z \cdot (X' + Y')$ lies in $\sigma(L\bar{s})$. Their sum lies in $\sigma(L\bar{s})$. Applying the outer σ:

$$P_1 + P_2 = \sigma(\text{element of } \sigma(L\bar{s})) \in \sigma(\sigma(L\bar{s})) = L\bar{s},$$

since $\sigma^2 = \mathrm{id}$. □

4.5. **Condition 3: the block sum lies in Ls.** This is the crux of the proof. The standard (untwisted) product fails here because $Ls \cdot Ls \not\subseteq Ls$ (Remark **??**). The twist redirects the computation through $\sigma(Ls)$, where closure holds.

Proposition 4.9. *For all $u, v \in \Lambda$, the block sum $P_1 + P_2 + P_3$ lies in Ls.*

Proof. From (**??**)–(**??**):

$$P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big).$$

Since $u \in \Lambda$, condition 3 gives $x + y + z \in Ls$, so

$$X + Y + Z = \sigma(x + y + z) \in \sigma(Ls).$$

Similarly $X' + Y' + Z' \in \sigma(Ls)$.

By Lemma D, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$, so

$$(X + Y + Z) \cdot (X' + Y' + Z') \in \sigma(Ls).$$

Applying the outer σ:

$$P_1 + P_2 + P_3 \in \sigma(\sigma(Ls)) = Ls.$$ □

4.6. **Why the standard product fails.**

Proposition 4.10. *The standard (untwisted) triple-octonion product $\star_{\mathrm{std}}$ (using $\cdot$ in place of $\cdot_\sigma$) satisfies Wilson's conditions 1 and 2 for all $u, v \in \Lambda$, but fails condition 3.*

Proof. Conditions 1 and 2 for $\star_{\mathrm{std}}$ follow from Lemmas A and B (with $\sigma = \mathrm{id}$, so $X = x$, etc.) and the fact that $L \cdot L\bar{s} \subseteq L\bar{s}$ (verified on $\mathbb{Z}$-bases: all 64 products check).

For condition 3, the block sum becomes $(x + y + z) \cdot (x' + y' + z')$ with both factors in Ls. This requires $Ls \cdot Ls \subseteq Ls$, which fails: the product of two explicit Ls-basis vectors yields an element outside Ls (Remark **??**). □

Remark 4.11. The mechanism is now clear. The twist introduces σ in two places: it maps the inputs through σ (sending Ls to $\sigma(Ls)$) and applies σ to the output (sending $\sigma(Ls)$ back to Ls). The standard product fails because Ls is not closed under multiplication. By Lemma E, $\sigma(Ls) \neq Ls$: the twist moves the condition-3 sublattice to a *different* sublattice of L that happens to be closed (Lemma D). The isomorphism between $(\mathbb{O}, \cdot)$ and $(\mathbb{O}, \cdot_\sigma)$ does not by itself explain closure — what matters is the interaction between the permutation and the specific sublattice geometry of Wilson's construction.

This completes the proof of Theorem 1.1. □

5. Algebraic properties

The order $(\Lambda, +, \star)$ has been tested for several standard algebraic properties. The results are summarised below.

Property	Holds?	Details
Multiplicative identity	No	(e_0, e_0, e_0) has $N = 3$; not in Λ
Commutativity	No	$< 0.1\%$ of pairs commute
Norm multiplicativity	No	$\approx 47\%$ of pairs satisfy $N(u \star v) = N(u)N(v)$
Left alternativity	No	$< 0.1\%$ of pairs satisfy $(x \star x) \star y = x \star (x \star y)$
Right alternativity	No	$< 0.1\%$ of pairs satisfy $(x \star y) \star y = x \star (y \star y)$
Flexibility	No	$\approx 8.6\%$ of pairs satisfy $(x \star y) \star x = x \star (y \star x)$
Power-associativity	No	$\approx 29\%$ of elements satisfy $x^2 \star x = x \star x^2$

Remark 5.1. The product norm on $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ takes values in $\{16, 32, 48, 64, 80, 96, 112, 128\}$, all multiples of 16. The mode is 64 (the norm-multiplicative value $N(u)N(v) = 8 \times 8$).

Remark 5.2. The order is non-unital: the ambient algebra $(\mathbb{R}^{24}, +, \star)$ has multiplicative identity (e_0, e_0, e_0), but this element has squared norm 3 and does not lie in Λ. Thus $(\Lambda, +, \star)$ is a non-unital order.

The order does not satisfy any of the classical algebraic identities (associativity, alternativity, flexibility, power-associativity, commutativity). The defining property is *closure*: the Leech lattice is stable under the product. The relationship between $\star$ and the Conway group Co_0 (the automorphism group of Λ as a lattice) is an open question.

6. Research methodology

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of 36 directed prompts, governed by a written methodology (the "Manifesto") established at the outset.

The human researcher (the author) brought the initial hypothesis — motivated by prior work on Okubo algebras and autotopies of normed composition algebras — established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work: implementing octonion algebras, Wilson's Leech lattice construction, trial algebras, and closure tests, and contributed mathematical analysis.

The computational verification was a vehicle in the discovery process, designed to mediate distrust into received knowledge and intermediate results. Tests were written from first principles to independently verify

every foundation: that octonion multiplication satisfies the composition property $N(xy) = N(x)N(y)$, that Wilson's construction produces a lattice with the correct minimal shell of 196,560 vectors, and that the E_8 lattice is indeed a maximal order. This independent verification is how we know that the properties stated in the literature are correctly implemented. Closure of the twisted product was confirmed on over 12,000,000 random pairs of minimal vectors with zero failures before the symbolic proof was found.

The symbolic proof (Section **??**) and the exact-arithmetic verifications of Lemmas C and D use Python's `fractions.Fraction` type, operating entirely in $\mathbb{Q}$ with no floating-point arithmetic. The $\mathbb{Z}$-basis checks involve solving 64 systems of 8 linear equations each, all yielding integer solutions.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology — understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach — was essential. The failure analysis of the untwisted product (exclusively condition 3, exclusively type-3×type-3) directly pointed to the mechanism that the twist needed to correct.

The AI agent violated its own logging protocol on three separate occasions (prompts 011, 018–019, and 029–031), each time caught by the human researcher through review. These violations and their corrections are part of the record.

The full record — all prompts, code, evidence, and git history — is available at the repository [10]. The reader can form their own assessment of what this means for questions of discovery and attribution.

7. Related work

Wilson (2009) [9] characterises Λ as $\{(x, y, z) \in L^3 : \text{conditions } 1\text{–}3\}$ and describes the Conway group using 3×3 octonionic matrices. This is the framework used in the present paper. Wilson does not define or test a multiplicative product on Λ.

Dixon (2010) [6] constructs Λ using the XY-product $A \circ B = (A \cdot l_0)(l_0^{-1} \cdot B)$, an alternative octonion multiplication. Dixon does not test or claim closure of Λ under the XY-product.

Baez and Egan (2014) [2] explore the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ as a framework for Λ. They find that $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$ is

closed under the doubled Jordan product $2(xy+yx)$ but not under the standard Jordan product $x \circ y = xy+yx$. Baez poses the open question of finding an "exceptional Leech–Jordan ring." This approach differs fundamentally from ours: the Jordan product is commutative and 27-dimensional, while ours is non-commutative and 24-dimensional.

Kirmse (1924) [7] exhibited the E_8 lattice as a unimodular integral quadratic lattice inside the real octonion algebra. His lattice is not closed under multiplication, an error corrected by Coxeter [5] using the index-doubling permutation. The idea of modifying an octonion multiplication to achieve lattice closure has its roots in this correction.

8. Conclusion

The Leech lattice admits an order under a remarkably simple construction: three identical copies of a transposition-twisted octonion algebra with $\mathbb{Z}_3$ routing. The symbolic proof (Section **??**) reduces closure to five lemmas, of which the crux is Lemma D: the sublattice $\sigma(Ls)$ — unlike Ls itself — is closed under the standard octonion product. The twist maps the condition-3 sublattice to this closed sublattice, and the involutory nature of σ maps the result back.

Several questions remain open:

- The automorphism group of $(\Lambda, +, \star)$ and its relationship to the Conway group Co_0.
- Whether the order is maximal in any appropriate sense.
- Classification of the algebra $(\mathbb{R}^{24}, +, \star)$: it is non-associative, non-alternative, non-flexible, and non-unital (as an order), placing it outside the classical families.
- Whether other permutations of the Fano triples (beyond transpositions) also yield closed products on Λ.

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